

(IC3) through resistor R5. Zener diode ZD1 regulates the supply to 2.2V required by IC3. 2.2V is fed to Vcc pin of BT8031, which outputs ding dong signal at its MO pin. This output tone cannot drive the speaker (LS1) directly. Therefore BC547 and BD678 transistors are used as amplifier stages.

Working of the circuit is simple. A touch on the touch plate is detected and a ding dong sound produced by the speaker immediately.

Construction and testing

An actual-size, single-side PCB for the ding dong tone generator circuit is shown in Fig. 2 and its component layout in Fig. 3. Mount the compo-

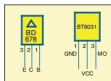


Fig. 4: Pin details of BT8031 IC

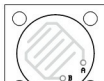


Fig. 5: Suggested touch plate

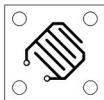


Fig. 6: Touch plate PCB

nents on the PCB. Use IC bases for IC1 and IC2. Solder the ding dong IC on the PCB. Pin details of BT8031 IC are shown in Fig. 4. The touch plate can be a piezoelectric plate taken out from a buzzer or it can be formed by two copper wires or two

copper tracks fabricated on a PCB as shown in Fig. 5. Points A and B are connecting terminals to the circuit. Preset VR1 is used to control the sensitivity of the touch plate.

To test the circuit for proper functioning, touch the touch plate to produce ding dong sound. If there is no output from the speaker, check the voltage status at various test points as per the table.

If the circuit is working well, enclose it in a suitable cabinet. Use a 9V PP3 battery or any 9V DC adaptor for powering the circuit. ●

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